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Ayush Agarwal

Technical Skills:

- **Languages:** Python, TypeScript
- **AI/ML:** Machine Learning, Deep Learning, NLP, LLMs, RAG
- **Libraries/Frameworks:** Scikit-learn, Pandas, NumPy, NLTK, spaCy, LangChain
- **Models/Techniques:** LDA, Random Forest, Decision Tree, AdaBoost, KNN, TF-IDF, NER, Embeddings (MPNet), Prompt Engineering
- **Retrieval/Graph:** FAISS, BM25, Knowledge Graphs, PageRank, RRF, Vector Databases
- **Backend:** FastAPI, REST APIs, WebSockets, SSE, Async Programming
- **Frontend:** React, Streamlit, Tailwind CSS
- **Databases:** SQLite (FTS5), Azure CosmosDB, Azure Blob Storage
- **Cloud/DevOps:** Microsoft Azure, Docker
- **AI Systems:** Multi-Agent Systems, Function Calling, MCP, Pydantic, Instructor
- **Tools:** Tesseract OCR, pdfplumber, pdf2image, PIL, NetworkX, PyVis, RapidFuzz, Joblib
- **Practices:** uv, Ruff, PEP8
- **Auth/Security:** OAuth 2.0, MSAL
- **Monitoring:** OpenTelemetry, Arize Phoenix

Certification:

- Prompt Engineering for Developers – DeepLearning.AI (Andrew Ng)
- Machine Learning – NPTEL
- Introduction to Large Language Models – Google Cloud
- Generative AI – Google Cloud

EDUCATION

Board	Tenure	Educational institution	CGPA/Percentage
B. Tech (CSE) Core	Sep 2023 - Ongoing	VIT Bhopal University	8.52/10
Class XII	May 2022	D.A.V Public School, Ranchi	88.0%
Class X	May 2020	D.A.V Nandraj Public School, Ranchi	91.0%

ACADEMIC PROJECTS

End-to-End AI System

Hybrid Legal RAG: Sparse–Dense–Graph Retrieval System

- Engineered a production-grade **retrieval-augmented system** combining BM25 (FTS5), dense embeddings (MPNet + FAISS), and graph-based ranking (PageRank)
- Implemented **Reciprocal Rank Fusion (RRF)** to unify multiple retrieval modalities for improved ranking quality
- Processed and indexed **26K+ documents**, enabling high-precision semantic and keyword-based search
- Achieved **82.4% Top-1 accuracy and NDCG@10 of 0.89**, outperforming single-modality baselines
- Built a **FastAPI backend** supporting search, conversational querying, and structured summarization
- Conducted **ablation studies** to validate robustness under high-noise conditions
- Containerized deployment using **Docker** with memory-efficient optimizations

Full-Stack LLM Application

PolarBrief AI – Legal Argument Analyzer

- Developed an **AI-powered document analysis tool** to extract, classify, and score arguments from unstructured PDFs
- Integrated **LLMs via LangChain** for summarization, heading generation, and structured scoring
- Implemented **argument polarity classification** (Pro/Con/Neutral) with weighted ranking using TF-IDF + LLM outputs
- Designed a **hybrid OCR + text extraction pipeline** using Tesseract and PDF parsers
- Generated structured outputs in **JSON/PDF formats** with citation-aware summaries

	• Built an interactive **Streamlit interface** for visualization and report generation
NLP Pipeline System	**Knowledge Graph Pipeline for Document Intelligence** • Built an end-to-end **knowledge graph extraction** pipeline from unstructured documents using NLP and LLMs • Implemented **multi-stage processing**: OCR, cleaning, chunking, entity-relation extraction, and validation • Designed **deterministic ID mapping and cross-chunk entity resolution** using fuzzy matching • Normalized entities and relations to construct a **clean, canonical knowledge graph** • Visualized graphs using **NetworkX and PyVis** for interactive exploration • Exported structured graph data for downstream analytics
Machine Learning Pipeline	**WESAD Multimodal Emotion Recognition** • Implemented and compared ML models (**LDA, Random Forest, Decision Tree, AdaBoost, KNN**) for emotion classification • Processed **multimodal physiological signals** (ECG, EDA, EMG, Respiration, Temperature, BVP, ACC) • Designed experiments to evaluate **sensor modality combinations** (chest vs wrist vs combined) • Built pipelines **for cross-validation, accuracy, and F1-score evaluation** • Generated detailed reports identifying **optimal modality configurations**

INTERNSHIP	
AI/ML Intern – Aug 2025 – Present	▪ **Docu3C Technologies:** **Reported to: Daniel Ingitaraj K, Head of Product Engineering** • Developed and deployed **AI/LLM-based applications** for financial document intelligence using Azure AI ecosystem • Built **end-to-end machine learning and LLM pipelines** with structured outputs using **Pydantic and schema validation** • Designed **multi-agent systems and tool-calling workflows** using **MCP (Model Context Protocol)** • Implemented **LLM orchestration pipelines** using **Instructor** for structured generation • Built scalable backend systems with **FastAPI**, supporting **real-time streaming (SSE/WebSockets)** and async execution • Developed **interactive UI applications (Streamlit/React)** for AI-driven workflows and validation systems • Leveraged **Azure ML, Cognitive Services, and GPU infrastructure** for scalable experimentation and deployment • Maintained **production-grade code quality** using **PEP8 standards, Ruff linting, and modular architecture** • Utilized modern tooling **(uv)** for efficient dependency and environment management • Conducted **benchmarking, evaluation, and iterative optimization** of AI systems

CO-CURRICULARS	
Coding	▪ Solved **125+ Data Structures & Algorithms problems** on LeetCode

EXTRA-CURRICULARS AND ACHIEVEMENTS	
Achievements	▪ Worked on **advanced AI/ML and LLM-based systems** including RAG pipelines, multi-agent systems, and document intelligence ▪ Gained hands-on experience with **Azure AI ecosystem and production deployments** ▪ Actively engaged in **collaborative development and agile workflows**
Responsibilities	▪ **Event Lead,** Linpack Club, VIT Bhopal University Organized and managed technical events and workshops, coordinating with teams and handling end-to-end execution Led planning and operations, ensuring smooth execution and participant engagement

	▪ **AI/ML Intern,** docu3C Collaborated in building scalable AI systems and participated in team-based engineering workflows
Extracurricular	▪ Volleyball (represented Thunders at Aavahan, VIT)

ADDITIONAL INFORMATION	
Hobbies	▪ Exploring AI advancements ▪ Volleyball
Languages	▪ English, Hindi